

MH1812 – Additional Exercises for Continuous Assessment 1

Q1: Prove that $\sqrt{3}$ is irrational.

Q2: Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Q3: Compute $5^{2015} \bmod 9$.

Q4: Compute $3^{3^{10}} \bmod 10$.¹

Q5: Suppose a is an integer and p is a prime number such that $p|a$ and $p|(a+3)$. What can you deduce about p ?

Q6: Prove by mathematical induction that, for $q \neq 1$ and integer $r \geq 0$,

$$a + aq + aq^2 + \cdots + aq^r = \frac{a(q^{r+1} - 1)}{q - 1}.$$

Q7: Let $a_k = a_1 + (k - 1) \cdot d$ for non-negative integer k . Prove by mathematical induction that

$$\sum_{i=1}^n a_i = n \cdot \frac{a_1 + a_n}{2}.$$

Q8: Prove by mathematical induction that, for any integer $n \geq 1$,

$$(n+1)(n+2) \cdots (2n-1)(2n) = 2^n \cdot 1 \cdot 3 \cdot 5 \cdots (2n-1),$$

Q9: Let p_n be the n -th prime number ($p_1 = 2$, $p_2 = 3$, $p_3 = 5$, ...). Show by mathematical induction that

$$p_{n+2} \geq 3n.$$

Q10: Show by mathematical induction that

$$2 \cdot 7^n \equiv 2^n \cdot (5n + 2) \pmod{25}.$$

¹In an iterated exponential, one calculates the exponent first, $a^{b^c} = a^{(b^c)}$, e.g., $4^{3^2} = 4^9 = 262144$.

Q11: Show $p \rightarrow (q \vee r) \equiv (p \wedge \neg q) \rightarrow r$

Q12: Show $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Q13: Verify the following argument:

$$\begin{array}{l} \neg p \vee q \rightarrow r \\ s \vee \neg q \\ \neg t \\ p \rightarrow t \\ \neg p \wedge r \rightarrow \neg s \\ \therefore \neg q \end{array}$$

Q14: Verify the following argument:

$$\begin{array}{l} p \vee q \\ q \rightarrow r \\ p \wedge s \rightarrow t \\ \neg r \\ \neg q \rightarrow u \wedge s \\ \therefore t \end{array}$$

Q15: Verify the following argument:

$$\begin{array}{l} \neg p \rightarrow r \wedge \neg s \\ t \rightarrow s \\ u \rightarrow \neg p \\ \neg w \\ u \vee w \\ \therefore \neg t \end{array}$$

Q16: Determine whether the following statements are true or false.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y = 0$.
2. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y = 0$.

Q17: Write a new statement of the following by interchanging $\forall$ and $\exists$ (e.g., $\forall x \exists y$ becomes $\exists y \forall x$), and state the truth of both versions before and after the interchange.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x < y$.
2. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}^-, x > y$. Here $\mathbb{R}^-$ denotes the set of negative reals.

Q18: Write the negation of the following statements.

1. For every real number x , if $x^2 \geq 1$ then $x > 0$.
2. For any integers a, b and c , if $a - b$ is even and $b - c$ is even, then $a - c$ is even.

Q19: Write the negation of the following sentences.

1. $\forall x \in D, \forall y \in E, P(x, y)$
2. $\exists x \in D, \exists y \in E, P(x, y)$
3. $\exists x \in D, \forall y \in E, \exists z \in F, \neg P(x, y, z) \rightarrow Q(x, y, z)$